

§ 138. Determination of Signs. Quadratic Residues

In the preceding paragraph we derived two formulas in which signs still remained to be determined, namely

$$2^{\frac{n-1}{2}} \prod_{\nu=1}^{(n-1)/2} \cos \frac{2\nu m \pi}{n} = \pm 1, \quad (1)$$

$$2^{\frac{n-1}{2}} \prod_{\nu=1}^{(n-1)/2} \sin \frac{2\nu m \pi}{n} = \pm \sqrt{n}, \quad (2)$$

where m is relatively prime to the odd number n , and ν runs through the sequence

$$(\nu) \quad 1, 2, \dots, \frac{n-1}{2}.$$

We shall retain the sign ν for the numbers of this sequence.

Every integer gives, on division by n , as remainder one of the numbers ν , or 0, or $n - \nu$. Instead of $n - \nu$ we may, if negative remainders are allowed, choose $-\nu$. Thus, if the numbers divisible by n are disregarded, one of the numbers $\pm \nu$ remains as remainder. We call this the absolutely least remainder, in distinction from the least remainder in the ordinary sense, which is taken from the numbers $1, 2, \dots, n - 1$.

Let m be relatively prime to n . Consider the sequence

$$(m\nu) \quad m, 2m, 3m, \dots, \frac{n-1}{2}m$$

and form for each term the absolutely least remainder ρ :

$$(\rho) \quad \pm \nu_1, \pm \nu_2, \pm \nu_3, \dots, \pm \nu_{\frac{1}{2}(n-1)}.$$

Among these numbers ρ no two equal numbers and no two opposite numbers occur. For if the sum or difference of two such numbers were zero, then for two different numbers ν, ν' from (ν) the number

$$m(\nu \pm \nu')$$

would be divisible by n , hence $\nu \pm \nu'$ itself would be divisible by n , which is impossible since both ν and ν' are less than $n : 2$. Thus, apart from sign and order, the totality of the numbers (ρ) agrees with the totality of the numbers (ν) .

In formulas (1) and (2), because of the periodicity of sine and cosine, we may replace νm by ρ ; and since $\cos(-\rho) = \cos \rho$, we may in formula (1) also replace $m\nu$ by ν . Hence the sign in (1) is independent of m . To determine it we use the fact that the cosine of an angle in the first quadrant is positive and in the second quadrant negative. Thus the sign in (1) is positive or negative according as an even or odd number of the angles

$$\frac{2\nu\pi}{n}$$

lies in the second quadrant, or equivalently according as an even or odd number of the numbers ν lies between $\frac{n}{4}$ and $\frac{n}{2}$.

If x is not an integer, write $E(x)$ for the greatest integer smaller than x , so that x lies between $E(x)$ and $E(x) + 1$. The number of integers lying between x and y is then $E(y) - E(x)$. We must therefore decide whether

$$E\left(\frac{n}{2}\right) - E\left(\frac{n}{4}\right)$$

is even or odd. The four cases are

$$\begin{aligned} n = 8k + 1, & \quad E(n/2) = 4k, & \quad E(n/4) = 2k, \\ n = 8k + 3, & \quad E(n/2) = 4k + 1, & \quad E(n/4) = 2k, \\ n = 8k + 5, & \quad E(n/2) = 4k + 2, & \quad E(n/4) = 2k + 1, \\ n = 8k + 7, & \quad E(n/2) = 4k + 3, & \quad E(n/4) = 2k + 1. \end{aligned}$$

Therefore the positive sign is to be taken in (1) when n is of the form $8k + 1$ or $8k + 7$, and the negative sign when n is of the form $8k + 3$ or $8k + 5$. In the first cases

$$\frac{n^2 - 1}{8}$$

is even, in the last two it is odd. Hence

$$2^{\frac{n-1}{2}} \prod_{\nu=1}^{(n-1)/2} \cos \frac{2\nu m \pi}{n} = (-1)^{\frac{n^2-1}{8}}. \quad (3)$$

In formula (2) the sign depends on m : it is positive or negative according as an even or odd number of negative signs occurs among the numbers (ρ) . To express the dependence of this sign on m and n we write (2) in the form

$$2^{\frac{n-1}{2}} \prod_{\nu=1}^{(n-1)/2} \sin \frac{2\nu m \pi}{n} = \left(\frac{m}{n}\right) \sqrt{n}, \quad (4)$$

where

$$\left(\frac{m}{n}\right) = \pm 1.$$

The symbol $\left(\frac{m}{n}\right)$ was introduced into number theory from another direction and in a more special form by Legendre, and was generalized by Jacobi. We shall call it the Legendre symbol. Its meaning for number theory will appear presently; first we have the following laws.

1. The value of $\left(\frac{m}{n}\right)$ is $+1$ or -1 according as the number of negative signs among the absolutely least remainders of $m\nu$ is even or odd.

For $m = 1$ the positive sign holds in (2), and hence

$$\left(\frac{1}{n}\right) = +1.$$

2. If m is replaced by $-m$, the sign of every factor on the left of (4) changes. Since the number of factors is $\frac{n-1}{2}$, we get

$$\left(\frac{-m}{n}\right) = (-1)^{\frac{n-1}{2}} \left(\frac{m}{n}\right), \quad (3')$$

and for $m = 1$

$$\left(\frac{-1}{n}\right) = (-1)^{\frac{n-1}{2}}. \quad (4')$$

3. If $2m$ is put for m , and the formula

$$\sin \frac{4\nu m \pi}{n} = 2 \sin \frac{2\nu m \pi}{n} \cos \frac{2\nu m \pi}{n}$$

is used, then (3) gives

$$\left(\frac{2m}{n}\right) = (-1)^{\frac{n^2-1}{8}} \left(\frac{m}{n}\right), \quad (5)$$

and for $m = 1$

$$\left(\frac{2}{n}\right) = (-1)^{\frac{n^2-1}{8}}. \quad (6)$$

4. If m is changed by a multiple of n , all factors of the product (4) remain unchanged; consequently

$$\left(\frac{m}{n}\right) = \left(\frac{m'}{n}\right), \quad \text{if } m \equiv m' \pmod{n}. \quad (7)$$

Now let not only n but also m be odd and positive. We apply formula (29) of § 137, with n there replaced by m :

$$\frac{\sin m\varphi}{\sin \varphi} = (-1)^{\frac{m-1}{2}} 2^{m-1} \prod_{\beta} (\sin^2 \varphi - \beta), \quad (5)$$

where β runs through all roots of $\Phi_m(x) = 0$ of the form

$$\beta = \sin^2 \frac{2\mu\pi}{m}. \quad (6)$$

If m' is relatively prime to n , then

$$\alpha = \sin^2 \frac{2\nu m'\pi}{n} \quad (7)$$

runs through all roots of $\Phi_n(x) = 0$. Putting in (5)

$$\varphi = \frac{2\nu m'\pi}{n} \quad (8)$$

and taking the product over all ν , we obtain

$$\frac{\prod_{\nu} \sin \frac{2\nu m m' \pi}{n}}{\prod_{\nu} \sin \frac{2\nu m' \pi}{n}} = (-1)^{\frac{(m-1)(n-1)}{4}} 2^{\frac{(m-1)(n-1)}{2}} \prod_{\alpha} \prod_{\beta} (\alpha - \beta). \quad (9)$$

The right hand side is wholly independent of m' . Thus the value of the left hand side is the same as for $m' = 1$. Substituting the values of the products from (4) gives

$$\left(\frac{mm'}{n}\right) : \left(\frac{m'}{n}\right) = \left(\frac{m}{n}\right) : \left(\frac{1}{n}\right),$$

or

$$\left(\frac{mm'}{n}\right) = \left(\frac{m}{n}\right) \left(\frac{m'}{n}\right).$$

At first m and m' were assumed positive and odd; but by the preceding laws the same result remains valid when m or m' is negative or even, provided only that both are relatively prime to n .

Assuming again that m is odd and positive, (4) and (9) give

$$\left(\frac{m}{n}\right) = (-1)^{\frac{(m-1)(n-1)}{4}} 2^{\frac{(m-1)(n-1)}{2}} \prod_{\alpha} \prod_{\beta} (\alpha - \beta). \quad (10)$$

If m and n are interchanged here, each factor in the double product on the right changes sign; everything else remains the same. The number of these factors is

$$\frac{n-1}{2} \cdot \frac{m-1}{2},$$

and therefore

$$\left(\frac{m}{n}\right) = (-1)^{\frac{(m-1)(n-1)}{4}} \left(\frac{n}{m}\right). \quad (9')$$

Equivalently, $\left(\frac{m}{n}\right)$ is equal to $\left(\frac{n}{m}\right)$ if at least one of the odd numbers m, n is of the form $4k+1$, and is the negative of $\left(\frac{n}{m}\right)$ if both are of the form $4k+3$.

This famous theorem is known as the law of reciprocity. The proof given here is due to Eisenstein. With its help, and using 5 and 6, one can determine the value of $\left(\frac{m}{n}\right)$ very quickly, proceeding as if one were computing the greatest common divisor of m and n .

From 8 and 9 follows one last theorem, valid when n and n' are positive odd numbers relatively prime to m :

$$\left(\frac{m}{n}\right) \left(\frac{m}{n'}\right) = \left(\frac{m}{nn'}\right). \quad (10')$$

Its correctness follows, for positive odd m , from the congruence

$$(nn' - 1) - (n - 1) - (n' - 1) = (n - 1)(n' - 1) \equiv 0 \pmod{4}, \quad (7')$$

after applying formula 9 and then 8 to both sides of 10; when m is even or negative it follows from 4 and 6.

These theorems reduce $\left(\frac{m}{n}\right)$ to a product of values $\left(\frac{q}{p}\right)$ with p, q prime. In practice, however, $\left(\frac{m}{n}\right)$ is more easily computed directly by 9, without this decomposition.

We shall now learn another meaning of $\left(\frac{m}{p}\right)$ when p is an odd prime. This is the meaning through which Legendre originally introduced the symbol. If all integers x not divisible by p are squared and their residues modulo p are examined, only $\frac{p-1}{2}$ distinct residues are obtained: congruent numbers give the same square residues, and opposite numbers also give the same square residues. All these residues are certainly obtained by squaring

$$(\nu) \quad 1, 2, 3, \dots, \frac{p-1}{2}.$$

These squares are all distinct modulo p , since for distinct ν, ν' in this range

$$\nu^2 - \nu'^2 = (\nu - \nu')(\nu + \nu')$$

cannot be divisible by p . Thus among

$$(s) \quad 1, 2, 3, \dots, p-1$$

one half occur as residues of x^2 , and the other half do not. The former numbers, together with all numbers congruent to them modulo p , are called the quadratic residues of p , the latter the quadratic non-residues of p .

If m is a quadratic residue, then the congruence

$$x^2 \equiv m \pmod{p}$$

is soluble, and by 7 and 8

$$\left(\frac{m}{p}\right) = \left(\frac{x^2}{p}\right) = \left(\frac{x}{p}\right)^2 = +1.$$

Conversely, the result from § 135 says that as m runs through $1, 2, \dots, p-1$, the symbol $\left(\frac{m}{p}\right)$ has the positive and negative signs equally often. Hence the negative sign occurs precisely for the quadratic non-residues. Thus:

11. If p is an odd prime and m is not divisible by p , then

$$\left(\frac{m}{p}\right) = +1 \quad \text{or} \quad -1$$

according as m is a quadratic residue or a quadratic non-residue of p .

This contains all the principal theorems on quadratic residues; for example: the product of two quadratic residues, or of two non-residues, is a residue, while the product of a residue and a non-residue is a non-residue.

Third Book. Algebraic Quantities

Thirteenth Section. Galois Theory

§ 139. The Concept of a Field

A system of numbers is called a number field if it is so complete and closed in itself that the four fundamental arithmetical operations, addition, subtraction, multiplication and division, performed on any numbers of the system, except division by zero, always lead to numbers belonging to the same system. This concept, which divides kinds of numbers from a natural point of view, was introduced by Dedekind. It is of the highest importance for algebra, and it is useful to have a significant and expressive name for it.

The concept of a number field can be extended to all quantities with which one can calculate according to the rules of the four species; in particular to rational functions of any variables.

We call a function field a system of functions of one or more variables with the property that the four arithmetical operations can be carried out in the system without restriction and always lead to a definite function of the same system, with the single exception of division by zero. The variables may be wholly independent of one another; but it is also possible that certain relations between them have been fixed and must be observed in calculation.

A function of a function field counts as zero only when it vanishes identically, that is, for all values of the variables under consideration.

Since for the moment we do not wish to restrict the discussion, we shall simply speak of fields, and call the objects with which the operations are to be carried out, whether numbers or functions, quantities or elements of the field.

A field is therefore a system of quantities complete enough that its quantities can be added, subtracted, multiplied and divided. If all quantities of one field are contained in a second field, the first is called a divisor of the second. If Ω is a divisor of Ω' , we shall also say that Ω' is a field over Ω .

Strictly speaking, the number zero alone forms a field by itself; for brevity we shall exclude this case once and for all. The nearest example is then the totality of rational numbers. This field is a divisor of every other field: every field contains a non-zero quantity ω , hence also $\omega : \omega = 1$, and therefore all integers and, by division, all fractions.

Other examples of number fields are the complex numbers $x + yi$, with $i = \sqrt{-1}$ and rational x, y ; the totality of real numbers; and the totality of all complex numbers $x + yi$. As an example of a function field one may take all rational functions of one variable, including the constants, where the constants are restricted to a given number field or are left unrestricted.

§ 140. Adjunction

Let a field of quantities be denoted by Ω . If a quantity α not contained in it is added, a new system

$$\Omega, \alpha$$

arises, but not yet a field. To obtain a field, one must add all quantities that can be derived by combining α with the numbers of Ω by the fundamental operations. Thus one obtains an enlarged field Ω' containing both α and Ω , and completely determined by them. We shall call it the field Ω, α .

This addition of a new quantity α to a field Ω is called adjoining, and one says that Ω' arises from Ω by adjunction of α .

To Ω' one may again adjoin a quantity α' and obtain a third field Ω'' containing Ω, α, α' . But one may also adjoin two or more numbers to Ω simultaneously and obtain the same field. One may also adjoin one field to another field and obtain a field containing the quantities of both. Adjoining to a field one of its divisors produces no new field.

For example, adjoining $i = \sqrt{-1}$ to the field $\Re$ of rational numbers gives the field $\mathfrak{C}$ of complex numbers $x + yi$ with rational x, y . Adjoining a variable u to $\Re$ gives the field of rational functions in u with rational coefficients; adjoining u and i to $\Re$ gives rational functions in u with coefficients in $\mathfrak{C}$.

A field of quantities is also often called, following Kronecker, a domain of rationality. This expression is useful when one wants to indicate that, in a given problem, the quantities of that field are to be considered known or rational.

§ 141. Functions in a Field

We shall often consider integral rational functions of one variable x ,

$$f(x) = a_0x^m + a_1x^{m-1} + \cdots + a_{m-1}x + a_m, \quad (1)$$

whose coefficients belong to a given field Ω . We shall call these functions contained in Ω , or simply functions in Ω . They are not to be confused with functions that belong to the field Ω when Ω itself is a function field.

Earlier we saw that such a function has m roots $\alpha_1, \dots, \alpha_m$ when the coefficients have fixed numerical values. If the coefficients are independent variables, the roots vary continuously with them as long as the coefficients remain in a sufficiently small region where the discriminant does not vanish.

Let Ω be any field and $f(x)$ a function in Ω . Then either $f(x)$ can be decomposed into factors of lower degree with coefficients in Ω , or it cannot. In the first case it is reducible, in the second irreducible, in Ω . If the field is clear, one often simply says reducible or irreducible.

A linear function is irreducible in every field. A product of several functions of the form (1) is reducible. A function irreducible in Ω may become reducible in an enlarged field; for instance, adjoining a root α of $f(x) = 0$ makes $f(x)$ have the linear factor $x - \alpha$.

- I. An irreducible function $f(x)$ can have no common divisor with another function $F(x)$ whose coefficients belong to the same field, unless $F(x)$ is divisible by $f(x)$.

Indeed, the greatest common divisor of $F(x)$ and $f(x)$ can be found by rational operations and therefore belongs to Ω . Since $f(x)$ has no divisor in Ω except itself and constants, this greatest common divisor is either constant or $f(x)$.

It follows that an irreducible function cannot have repeated factors, for then it would have a common divisor with its derivative $f'(x)$.

- II. If $f(x)$ is irreducible and $F(x)$ vanishes for one root of $f(x) = 0$, then it vanishes for all roots of $f(x) = 0$.
- III. If $F(x)$ is of lower degree than the irreducible function $f(x)$ and vanishes for one root of $f(x)$, then $F(x)$ must vanish identically.

Integral rational functions of several variables $f(x, y, z, \dots)$ with coefficients in Ω are also called functions in Ω . They are reducible or irreducible according as they can or cannot be decomposed into factors that are themselves functions in Ω and actually contain some of the variables. For several variables one must distinguish between functions that are not decomposable at all, functions decomposable only with coefficients outside Ω , and functions decomposable into functions in Ω ; only the last kind will be called reducible in Ω .

The fundamental theorem on irreducible functions established earlier may now be stated in this setting:

- IV. A reducible function in Ω can be decomposed into irreducible factors in only one way, if functions differing only by a constant factor are not regarded as different.

§ 142. Algebraic Fields

Let Ω be a field and $F(x)$ a function in Ω . The equation

$$F(x) = 0 \quad (1)$$

is called an equation in Ω . It is reducible or irreducible according as $F(x)$ is reducible or irreducible. By adjoining to Ω a root α of such an equation, if α is not already in Ω , one obtains a new field $\Omega(\alpha)$, called an algebraic field over Ω .

If $\beta, \gamma, \dots$ are roots of other equations in Ω , or other roots of the same equation, simultaneous adjunction gives fields

$$\Omega(\alpha, \beta, \gamma, \dots).$$

This apparent enlargement of the concept will soon be reduced to adjunction of a single element.

The equation (1) with root α may be reducible. Among the irreducible factors of $F(x)$ there is one, denoted $f(x)$, that vanishes at $x = \alpha$. Taking its leading coefficient to be 1, it is uniquely determined. Thus

$$x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n = 0, \quad (2)$$

with $a_i \in \Omega$. The degree n of this equation is also called the degree of the field $\Omega(\alpha)$.

Every quantity Θ of $\Omega(\alpha)$ is obtained from α and quantities in Ω by the four operations; hence it can be written as a rational function of α ,

$$\Theta = \frac{\chi(\alpha)}{\psi(\alpha)}, \quad (3)$$

where $\chi(x)$ and $\psi(x)$ are functions in Ω . Since $\psi(\alpha) \neq 0$, $\psi(x)$ is not divisible by $f(x)$; and since $f(x)$ is irreducible, $\psi(x)$ is relatively prime to $f(x)$. Therefore functions $Q(x)$ and $\varphi(x)$ in Ω , with φ of degree at most $n - 1$, may be chosen so that

$$Q(x)f(x) + \varphi(x)\psi(x) = \chi(x). \quad (4)$$

Putting $x = \alpha$ gives

$$\frac{\chi(\alpha)}{\psi(\alpha)} = \varphi(\alpha). \quad (5)$$

Thus every element of $\Omega(\alpha)$ has one and only one representation

$$\Theta = c_0 + c_1\alpha + c_2\alpha^2 + \dots + c_{n-1}\alpha^{n-1}$$

with $c_i \in \Omega$. Uniqueness follows because the difference of two such expressions is a function of degree lower than n vanishing at α , and therefore is identically zero.

When the quantities of Ω are the given rational quantities and $\Omega(\alpha)$ is derived from them, we may say simply that $\Omega(\alpha)$ consists of all rational functions of α .

§ 143. Multiple Adjunction

Before investigating $\Omega(\alpha)$ more closely, we use a simple but important lemma.

If $\Phi_1(x, y, z, \dots), \Phi_2(x, y, z, \dots), \dots$ are integral rational functions whose coefficients are arbitrary and which do not all vanish identically, then rational numerical values can be assigned to the variables in infinitely many ways so that none of the functions Φ_i vanishes.

For one variable this is immediate. The general case follows by induction: write the functions as polynomials in one variable, choose the remaining variables so that not all coefficients vanish, and then choose the last variable to avoid the finitely many zeros.

Now let $\alpha, \beta, \gamma, \dots$ be algebraic quantities satisfying equations

$$A(x) = 0, \quad B(x) = 0, \quad C(x) = 0, \dots \quad (1)$$

in Ω , with no repeated roots. Every element of $\Omega(\alpha, \beta, \gamma, \dots)$ is an integral rational function in $\alpha, \beta, \gamma, \dots$ with coefficients in Ω .

Form a linear function

$$\xi = x\alpha + y\beta + z\gamma + \dots$$

If $\alpha', \beta', \gamma', \dots$ is any other choice of one root from each equation, put

$$\xi' = x\alpha' + y\beta' + z\gamma' + \dots,$$

and similarly form $\xi'', \xi''', \dots$. If the degrees of $A, B, C, \dots$ are $a, b, c, \dots$, the number of these quantities is

$$m = abc \dots$$

The differences $\xi - \xi', \xi - \xi'',$ and so on are non-identically-zero linear functions of $x, y, z, \dots$. By the lemma we may choose rational values of these variables so that the m quantities $\xi, \xi', \xi'', \dots$ are all distinct.

The elementary symmetric functions of these ξ 's are symmetric in the roots of each of the equations (1), and so are rationally expressible in their coefficients, hence belong to Ω . Thus

$$F(t) = (t - \xi)(t - \xi')(t - \xi'') \dots \quad (2)$$

is an equation in Ω without repeated roots and with one root ξ .

If $\Theta \in \Omega(\alpha, \beta, \gamma, \dots)$, let $\Theta', \Theta'', \dots$ be the corresponding values obtained by replacing the roots by $\alpha', \beta', \gamma', \dots$ and so on. Then

$$F(t) \left(\frac{\Theta}{t - \xi} + \frac{\Theta'}{t - \xi'} + \frac{\Theta''}{t - \xi''} + \dots \right) = \psi(t)$$

is an integral rational function of degree $m - 1$ in t with coefficients in Ω . Putting $t = \xi$ gives

$$\Theta = \frac{\psi(\xi)}{F'(\xi)}. \quad (3)$$

Therefore every Θ belongs to $\Omega(\xi)$, while every element of $\Omega(\xi)$ plainly belongs to $\Omega(\alpha, \beta, \gamma, \dots)$. Hence

$$\Omega(\xi) = \Omega(\alpha, \beta, \gamma, \dots).$$

§ 144. Primitive and Imprimitive Fields

We may now restrict attention to fields $\Omega(\alpha)$ generated by one root. Let

$$F(x) = 0 \quad (1)$$

be the relevant equation without repeated factors, of degree m , with roots

$$\alpha, \alpha_1, \alpha_2, \dots, \alpha_{m-1}. \quad (2)$$

If $F(x)$ is reducible in Ω , there is exactly one irreducible factor $f(x)$ of $F(x)$ for which α is a root:

$$f(x) = 0. \quad (3)$$

Let its degree be n , and let its roots be

$$\alpha, \alpha_1, \alpha_2, \dots, \alpha_{n-1}. \quad (4)$$

Then the degree of $\Omega(\alpha)$ is n . The fields

$$\Omega(\alpha), \Omega(\alpha_1), \dots, \Omega(\alpha_{n-1}) \quad (5)$$

are called the fields conjugate to $\Omega(\alpha)$.

Every $\Theta \in \Omega(\alpha)$ can be written as $\Theta = \chi(\alpha)$, with χ a polynomial in Ω of degree at most $n - 1$. Substituting the other roots gives

$$\Theta = \chi(\alpha), \quad \Theta_1 = \chi(\alpha_1), \quad \dots, \quad \Theta_{n-1} = \chi(\alpha_{n-1}), \quad (6)$$

the conjugates of Θ . Every symmetric function of these conjugates belongs to Ω . In particular, their sum

$$S(\Theta) = \Theta + \Theta_1 + \Theta_2 + \dots + \Theta_{n-1} \quad (7)$$

is called the trace of Θ , and their product

$$N(\Theta) = \Theta \Theta_1 \Theta_2 \dots \Theta_{n-1} \quad (8)$$

is called the norm of Θ . Conjugate quantities have the same traces and norms.

The product

$$(t - \Theta)(t - \Theta_1) \dots (t - \Theta_{n-1}) = \Phi(t) \quad (9)$$

is a polynomial of degree n in Ω , with roots Θ and its conjugates. Therefore every element of $\Omega(\alpha)$ is a root of an equation of degree n in Ω , whose other roots are its conjugates.

If $\Phi(t)$ is reducible, let $\varphi(t)$ be one irreducible factor. If $\varphi(\Theta) = 0$, then $\varphi[\chi(x)]$ and $f(x)$ have a common root. Since f is irreducible, $\varphi[\chi(x)]$ vanishes for every root α_i . Thus either all conjugates of Θ are distinct, in which case Φ itself is irreducible, or the distinct conjugates split into n_1 values each repeated n_2 times:

$$\varphi(t) = (t - \Theta)(t - \Theta_1) \dots (t - \Theta_{n_1-1}), \quad (10)$$

$$\Phi(t) = \varphi(t)^{n_2}, \quad (11)$$

$$n = n_1 n_2. \quad (12)$$

Thus $\Phi(t)$ is either irreducible or a power of an irreducible polynomial.

An element Θ of $\Omega(\alpha)$ whose conjugates are all distinct, and which therefore satisfies an irreducible equation of degree n , is called a primitive element of the field. Infinitely many such primitive elements can be found.

Every element ω of $\Omega(\alpha)$ can be rationally expressed in terms of any primitive element Θ . Indeed, if $\omega, \omega_1, \dots, \omega_{n-1}$ and $\Theta, \Theta_1, \dots, \Theta_{n-1}$ are the corresponding conjugates, then

$$\Phi(t) = (t - \Theta)(t - \Theta_1) \dots (t - \Theta_{n-1})$$

and

$$\Phi(t) \left(\frac{\omega}{t - \Theta} + \frac{\omega_1}{t - \Theta_1} + \dots + \frac{\omega_{n-1}}{t - \Theta_{n-1}} \right) = \Psi(t)$$

with $\Psi(t) \in \Omega[t]$. Setting $t = \Theta$ gives

$$\omega = \frac{\Psi(\Theta)}{\Phi'(\Theta)}.$$

Hence $\Omega(\Theta) = \Omega(\alpha)$.

If Θ is not primitive, not every element of $\Omega(\alpha)$ can be expressed rationally in terms of Θ . Then $\Omega(\Theta)$ is a divisor of $\Omega(\alpha)$, and its degree divides the degree of $\Omega(\alpha)$.

The field $\Omega(\alpha)$ is called primitive if it contains no imprimitive quantities apart from the quantities of Ω ; otherwise it is imprimitive. A field of prime degree is necessarily primitive. More generally, every divisor of $\Omega(\alpha)$ that contains Ω is an algebraic field $\Omega(\beta)$ over Ω , its degree divides the degree of $\Omega(\alpha)$, and equal degrees give identical fields. Thus an algebraic field over Ω is primitive precisely when, apart from Ω and itself, it has no divisor field over Ω .

§ 145. Normal Fields. Galois Resolvents

If $\Omega(\alpha)$ is an algebraic field, then the conjugate fields

$$\Omega(\alpha), \Omega(\alpha_1), \dots, \Omega(\alpha_{m-1})$$

all have the same degree; hence if one is contained in another, the two are identical. A field identical with all its conjugate fields is called a normal field. Such fields are also called Galois fields.

If a field $\Omega(\rho)$ of degree μ has the property that all numbers $\rho_1, \rho_2, \dots, \rho_{\mu-1}$ conjugate to ρ are contained in $\Omega(\rho)$, then it is a normal field.

An equation is called a normal equation if it is irreducible and all its roots can be rationally expressed in Ω by means of any one of them. A primitive element of a normal field of degree μ is a root of a normal equation of degree μ , and conversely the field generated by a root of a normal equation is a normal field of the same degree.

Every algebraic field can be led back to a normal field as follows. Let

$$F(x) = 0 \tag{1}$$

be any equation in Ω of degree m , reducible or irreducible, but without repeated roots, and let its roots be

$$\alpha, \alpha_1, \dots, \alpha_{m-1}. \tag{2}$$

The field

$$N = \Omega(\alpha, \alpha_1, \dots, \alpha_{m-1}) \tag{4}$$

generated by all roots is called the Galois field of $F(x) = 0$. If $F(x)$ is irreducible, N is also called the norm of any of the conjugate fields $\Omega(\alpha_i)$.

Choose a primitive element ρ of N and write $N = \Omega(\rho)$. Let μ be the degree of N , and let

$$g(t) = 0 \tag{5}$$

be the irreducible equation of degree μ satisfied by ρ . Since ρ is a rational function of the roots $\alpha, \alpha_1, \dots, \alpha_{m-1}$, permuting these roots gives quantities $\rho, \rho', \rho'', \dots$. The product

$$G(t) = (t - \rho)(t - \rho')(t - \rho'') \cdots$$

has coefficients symmetric in the roots of $F(x)$ and therefore in Ω . Since $G(t)$ and $g(t)$ have a common root and $g(t)$ is irreducible, $g(t)$ divides $G(t)$; hence all roots of $g(t)$ are contained in N . Thus N is a normal field.

A Galois resolvent of $F(x) = 0$ is an irreducible equation $g(t) = 0$ such that all roots of F can be rationally expressed by one root ρ of g , and one root of g can be rationally expressed by the roots of F . Every Galois resolvent is a normal equation, and every normal equation is its own Galois resolvent.

§ 146. The Substitutions of a Normal Field

Let $\Omega(\rho)$ be a normal field of degree μ and ρ a primitive element. Let

$$g(t) = 0 \quad (1)$$

be the irreducible equation of degree μ with one root $t = \rho$. The elements conjugate to ρ are

$$\rho, \rho_1, \rho_2, \dots, \rho_{\mu-1}. \quad (2)$$

Since all of them lie in the normal field, there are polynomials $\Theta_i(t)$ in Ω , of degree at most $\mu - 1$, such that

$$\rho_1 = \Theta_1(\rho), \quad \rho_2 = \Theta_2(\rho), \quad \dots, \quad \rho_{\mu-1} = \Theta_{\mu-1}(\rho). \quad (3)$$

If ω is any element of $\Omega(\rho)$, its conjugates have the form

$$\omega = \varphi(\rho), \quad \omega_1 = \varphi(\rho_1), \dots, \quad \omega_{\mu-1} = \varphi(\rho_{\mu-1}). \quad (4)$$

Because $g(t)$ is irreducible, if a rational function $\Phi(t)$ in Ω has one root in common with $g(t)$, then

$$\Phi(\rho), \Phi(\rho_1), \dots, \Phi(\rho_{\mu-1})$$

all vanish.

Now replace ρ in one of the functions $\Theta_k(t)$ by another root ρ_h . The result is again one of the roots. Indeed the set

$$\rho_h, \Theta_1(\rho_h), \Theta_2(\rho_h), \dots, \Theta_{\mu-1}(\rho_h) \quad (5)$$

coincides, apart from order, with

$$\rho, \Theta_1(\rho), \Theta_2(\rho), \dots, \Theta_{\mu-1}(\rho). \quad (6)$$

If two members of (5) were equal, then by the preceding principle the corresponding two members of (6) would be equal, which is impossible.

Thus, when ρ is replaced by ρ_h , every quantity conjugate to ρ passes to a definite other such quantity, and no two different quantities pass to the same one.

Replacing ρ in all rational functions of ρ by another root ρ_a defines a substitution, denoted

$$\delta_a = (\rho, \rho_a), \quad a = 0, 1, 2, \dots, \mu - 1.$$

The substitution δ_0 is the identity. If $\rho_h = \Theta_h(\rho)$, then δ_a carries ρ_h to

$$\rho_k = \Theta_h(\rho_a).$$

Likewise an arbitrary element $\omega = \varphi(\rho)$ is carried to $\omega_a = \varphi(\rho_a)$. The same substitution may therefore be written

$$\delta_a = (\rho_h, \rho_k) = (\rho_h, \Theta_h \Theta_a(\rho)). \quad (9)$$

Here ρ_h may be any root of $g(t)$, and the corresponding ρ_k is then determined by δ_a . Conversely either root in such a pair may be prescribed and the other is then determined.

Thus there are exactly μ distinct substitutions of $\Omega(\rho)$, including the identity. Each carries the totality of elements of the field into itself, every element going to a definite element and no two different elements going to the same one.

An element $\omega = \varphi(\rho)$ is said to allow or admit the substitution (ρ, ρ_a) , or δ_a , if

$$\varphi(\rho) = \varphi(\rho_a).$$

Elements admitting no substitution other than the identity are the primitive elements of the field. An element admitting all substitutions δ_a is necessarily an element of Ω , for it is identical with all its conjugates and so satisfies an equation of first degree in Ω .

§ 147. Composition of Substitutions

If in a function of ρ we first replace ρ by ρ_a and then ρ_a by ρ_b , the result is the same as if ρ had from the beginning been replaced by ρ_b . Thus if

$$\delta = (\rho, \rho_a), \quad \delta' = (\rho_a, \rho_b), \quad \delta'' = (\rho, \rho_b), \quad (1)$$

then carrying out δ and then δ' has the same effect as carrying out δ'' once. We therefore call δ'' the composition of δ and δ' and write

$$\delta\delta' = \delta''. \quad (2)$$

Of the three substitutions connected by $\delta\delta' = \delta''$, any two may be given arbitrarily, and the third is then uniquely determined.

The notation is multiplicative, but the order of the factors is essential: in general $\delta\delta'$ and $\delta'\delta$ need not be the same. The commutative law need not hold, but the associative law does:

$$(\delta\delta')\delta'' = \delta(\delta'\delta''). \quad (3)$$

Indeed choose roots so that

$$\delta = (\rho, \rho_a), \quad \delta' = (\rho_a, \rho_b), \quad \delta'' = (\rho_b, \rho_c). \quad (4)$$

Then both $(\delta\delta')\delta''$ and $\delta(\delta'\delta'')$ carry ρ to ρ_c . Hence both are (ρ, ρ_c) .

We therefore denote the composition of three substitutions by $\delta\delta'\delta''$, and similarly compose any number of adjacent components until the symbolic product has contracted to a single substitution. For example

$$(\rho, \rho_a)(\rho_a, \rho_b)(\rho_b, \rho_c)(\rho_c, \rho_e) = (\rho, \rho_e),$$

and the same argument extends to any number of components.